Chapter 1 • Real Numbers

Solved Exercise 1.2

1(i). Rationalize $\frac{13}{4+\sqrt{3}}$.

$\frac{13}{4+\sqrt{3}} \times \frac{4-\sqrt{3}}{4-\sqrt{3}}$

$= \frac{13(4-\sqrt{3})}{16-3}$

$4-\sqrt{3}$.

1(ii). Rationalize $\frac{\sqrt{2}+\sqrt{5}}{\sqrt{3}}$

$\frac{\sqrt{2}+\sqrt{5}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{6}+\sqrt{15}}{3}$

1(iii). Rationalize $\frac{\sqrt{2}-1}{\sqrt{5}}$

$\frac{\sqrt{2}-1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{10}-\sqrt{5}}{5}$

1(iv). Rationalize $\frac{6-4\sqrt{2}}{6+4\sqrt{2}}$.

$= \frac{68-48\sqrt{2}}{36-32}$

$= \frac{68-48\sqrt{2}}{4}$

$17-12\sqrt{2}$.

1(v). Rationalize $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$.

$= \frac{3+2-2\sqrt{6}}{3-2}$

$5-2\sqrt{6}$.

1(vi). Rationalize $\frac{4\sqrt{3}}{\sqrt{7}+\sqrt{5}}$.

$\frac{4\sqrt{3}(\sqrt{7}-\sqrt{5})}{7-5}$

$2\sqrt{3}(\sqrt{7}-\sqrt{5}) = 2\sqrt{21}-2\sqrt{15}$.

2(i). Simplify $(81/16)^{\frac{-3}{4}}$.

$(\frac{16}{81})^{\frac{3}{4}} = (\frac{2}{3})^3$

$\frac{8}{27}$

2(ii). Simplify $(3/4)^{-2} \div (\frac{4}{9})^3 \times \frac{16}{27}$

$= (\frac{4}{3})^2 \times (\frac{9}{4})^3 \times \frac{16}{27}$

$= (\frac{16}{9}) \times \frac{729}{64} \times (\frac{16}{27})$

12.

2(iii). Simplify $(0.027)^{\frac{-1}{3}}$.

$(\frac{27}{1000})^{\frac{-1}{3}} = (\frac{1000}{27})^{\frac{1}{3}}$

$\frac{10}{3}$

2(iv). Simplify the seventh root of $(x^{14}y^{21}z^{35})/(y^{14}z^7)$.

Inside the root: $x^{14}y^7z^{28}$.

Taking the seventh root divides each exponent by 7.

x^2yz^4.

2(v). Simplify $[5(25)^{n+1} - 25(5)^{2n}]/[5(5)^{2n+3} - (25)^{n+1}]$.

Numerator = $5^{2n+2}(5-1)$

denominator = $5^{2n+2}(25-1)$.

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Common factor cancels: $\frac{4}{24}$

$\frac{1}{6}$

2(vi). Simplify $[16^{x+1} + 20(4^{2x})]/[2^{x-3} \times 8^{x+2}]$.

Numerator = $2^{4x}(16 + 20)$

$= 36 \times 2^{4x}$.

Denominator = $2^{x-3+3x+6}$

$= 8 \times 2^{4x}$.

$\frac{36}{8} = \frac{9}{2}$

2(vii). Simplify $64^{\frac{-2}{3}} \div 9^{\frac{-3}{2}}$.

$64^{\frac{-2}{3}} = \frac{1}{16}$ and $9^{\frac{-3}{2}} = \frac{1}{27}$

$(\frac{1}{16}) \div \frac{1}{27}$

$\frac{27}{16}$

2(viii). Simplify $[3^n \times 9^{n+1}]/[3^{n-1} \times 9^{n-1}]$.

Exponent of 3 = n + 2(n+1) − [(n−1) + 2(n−1)]

= 5.

$3^5 = 243$.

2(ix). Simplify $[5^{n+3} - 6 \cdot 5^{n+1}]/[9 \cdot 5^n - 4 \cdot 5^n]$.

Numerator = $5^{n+1}(25 - 6)$

$= 19 \cdot 5^{n+1}$

denominator = $5 \cdot 5^n$

$= 5^{n+1}$.

19.

3. If $x = 3 + \sqrt{8}$, find the six requested expressions.

$\frac{1}{x} = \frac{3 - \sqrt{8}}{9 - 8}$

$= 3 - \sqrt{8}$.

(i) $x + \frac{1}{x} = 6$

(ii) $x - \frac{1}{x} = 2\sqrt{8}$.

(iii) $x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2$

$= 36 - 2$

$= 34$.

(iv) $x^2 - \frac{1}{x^2} = (x + \frac{1}{x})(x - \frac{1}{x})$

$= 12\sqrt{8}$.

(v) $x^4 + \frac{1}{x^4} = 34^2 - 2$

$= 1154$

(vi) $(x - 1/x)^2 = (2\sqrt{8})^2$

$= 32$.

4. Find rational p and q such that $\frac{8 - 3\sqrt{2}}{4 + 3\sqrt{2}} = p + q\sqrt{2}$.

$[(8 - 3\sqrt{2})(4 - 3\sqrt{2})]/(16 - 18) = (50 - 36\sqrt{2})/(-2)$

$= -25 + 18\sqrt{2}$.

$p = -25, q = 18$.

5(i). Simplify $[25^{\frac{3}{2}} \times 243^{\frac{3}{5}}]/[16^{\frac{5}{4}} \times 8^{\frac{4}{3}}]$.

$= \frac{125 \times 27}{32 \times 16}$

$\frac{3375}{512}$

5(ii). Simplify $54 \cdot \sqrt[3]{(27^{2x})}/[9^{x+1} + 216(3^{2x-1})]$.

Numerator = $54 \cdot 3^{2x}$.

Denominator = $9 \cdot 3^{2x} + 72 \cdot 3^{2x}$

$= 81 \cdot 3^{2x}$.

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$\frac{54}{81} = \frac{2}{3}$

5(iii). Simplify $\sqrt{([216^{\frac{2}{3}} \times 25^{\frac{1}{2}}]/(0.04)^{\frac{-3}{2}})}$.

$(0.04)^{\frac{-3}{2}} = 125$; inside the root $= \frac{36 \times 5}{125} = \frac{36}{25}$

$\frac{6}{5}$

5(iv). Simplify $(a^{\frac{1}{3}} + b^{\frac{2}{3}})(a^{\frac{2}{3}} - a^{\frac{1}{3}}b^{\frac{2}{3}} + b^{\frac{4}{3}})$.

Use $(u + v)(u^2 - uv + v^2) = u^3 + v^3$ with $u = a^{\frac{1}{3}}$, $v = b^{\frac{2}{3}}$.

$a + b^2$.

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